

1608 THE TELESCOPE

Hans Lippershey was a German lensmaker working in the Netherlands, then the center of the optical industry. He is believed to have invented the refracting telescope in 1608, though Zacharias Janssen may also have had a hand in its development. A refracting telescope uses two lenses to gather and focus light, making distant objects appear closer than they are. Lippershey's invention could make objects seem three times larger. It is likely that he intended his telescope for use at sea or on the battlefield.



Lippershey experiments with lenses



Galileo's telescope

Based on reports of Lippershey's telescope, Galileo built his own telescope in 1609. It could make distant objects appear eight times larger. He later built a telescope that could magnify objects 30 times.

The tube is made of strips of wood covered with leather.

1605

1610

1604

Falling objects

Galileo worked out a law that describes how objects fall under the influence of gravity. At the time, most scientists believed Greek philosopher Aristotle's idea, that the heavier an object is, the faster it falls. Galileo had already realized that all objects should fall at the same rate and land together, but that air resistance affects the falls of some objects more than others. He supposedly tested his idea by dropping cannonballs of various weights from the Leaning Tower of Pisa, Italy, although historians dispute the story.



Galileo dropping balls from the Leaning Tower of Pisa

Faster firing

The flintlock was first used in France in 1608. The spring-operated mechanism increased the firing rate and safety of handheld muskets and pistols. It remained in use for more than 200 years.



The falling flint strikes a spark to ignite the gunpowder.

1609

Planetary motion

German astronomer Johannes Kepler published a work confirming Copernicus's belief in a Sun-centered cosmos. He also used mathematics to prove that the planets travel in elliptical orbits around the Sun, and that their speed is not constant, they speed up when they come closest to the Sun.

